LAB2 Technical report

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CS353: Introduction to AI Robotics

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In this lab, students were expected to simulate a wall-following algorithm in Gazebo and then run it on the physical robot. The first key lesson we took away from this lab was the importance of taking precautions when designing robots and writing code. Examples of these precautions include testing code in a virtual environment and ensuring something is placed below the robot, so its wheels are elevated, and it doesn't fall off the table.

This lab was built on the previous lab, and the first tutorial did a great job of reinforcing the concepts of URDF and Gazebo. It was also implemented in a ROS version very similar to ours. It was very easy to understand and follow. The first major roadblock we faced as a team, besides the initial setup and deciding when to push, was lidar. It took days to get it working reliably, and at some point, we had to scrap the robot description package we had and start again. Another issue was that when we finished our controller bit and had it moving with the teleop twist keyboard, we were confused for a while about ROS2Control and Gazebo control. We attempted to switch to ros2control, but it did not work as expected.

This lab helped build our team's confidence. We learnt how to work collaboratively on GitHub, how to clean our repositories when someone pushes code, and how to pull it. We also reinforced our skills in creating Xacro, URDF, and launch files. Some areas we still need to improve as a team include figuring out the best way to split the workload. Initially, we were all working on the robot description package in case one or more of us couldn't do it. Another area is communication and hopefully we can improve on that in the next lab